Sprint 4 - Team 11

Error budget

Two targets, how much failure each one allows, and what we do when we go over.

	Trip cost accuracy	Backup service availability
Indicator	Difference between the estimated trip cost and the actual trip cost.	Availability of backup functionality when a primary third-party travel service or API is unavailable.
Target	At least 90% of trip estimates are within ±10% of the actual trip cost.	Backup functionality is available at least 99% of the time during third-party service outages.
Window	Monthly, using completed trips with available actual-cost data.	Monthly.
Who measures	Application analytics and the development team, by comparing estimated and actual trip costs.	Automated API monitoring and routine system checks.
Error budget	Up to 10% of trip estimates may fall outside the ±10% range.	Up to 1% downtime per month. In a 30-day month, about 7.2 hours.
What spends the budget	A trip estimate that differs from the actual trip cost by more than 10%.	Any time backup functionality is unavailable while a primary third-party service or API is down.
Policy action	If more than 10% of estimates miss the target, pause changes to the estimation feature and review the cost-calculation formula and data sources.	If backup downtime exceeds the 1% budget, prioritize reliability work, review backup providers, and add or improve fallback options.
Owner	Lead Developer	Project Manager / Technical Lead
Who we tell, and when	Notify the project team when the budget is exceeded and include the corrective action in the next project update.	Notify the project team immediately. If the outage affects users or a project deadline, notify the instructor or stakeholder in the next required update.

How the budgets are calculated

Trip cost accuracy
SLO = 90%
Error budget = 100% - 90% = 10%
Up to 10% of trip estimates may fall outside the ±10% range before the budget is spent.

Backup service availability
SLO = 99%
Error budget = 100% - 99% = 1%
30 days × 24 hours = 720 hours; 1% of 720 hours = 7.2 hours
Backup may be unavailable for up to about 7.2 hours per month before the budget is spent.